Tier 1 Reliability Assessment: AWS

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Selected cloud: Amazon Web Services (AWS)

AWS is the selected cloud for every task, service and runbook in this submission; no other cloud provider is used. No hybrid or second-cloud dependency exists. The non-AWS components are open-source software running inside the AWS workload on Amazon EKS, or operational tools rather than hosting dependencies, each justified where it appears: Istio service mesh (Task 3, chosen because AWS App Mesh reached end of support), KEDA for consumer-lag scaling and Fluent Bit for log shipping (Tasks 2 and 3), and the organization's existing paging and chat tools for on-call and incident channels (Tasks 2 and 5).

Summary

The service detects failures late because alerts watch components, not customer journeys. It turns provider slowdowns into regional incidents because retries are unbounded and stacked across layers. Its recovery is unproven because DR has never run end to end, and it lacks an owner because no team holds an error budget. Technical controls without ownership decay, and ownership without controls cannot stop a retry storm, so this plan fixes both:

- **Detect:** measure three customer journeys at the edge against 99.99% and page only on error budget burn, not on causes.
- **Contain:** allow retries at exactly one layer under a 20% retry budget, add circuit breakers, bulkheads and load shedding, and put both providers behind fallbacks.
- **Recover:** run a warm standby Region driven by an Amazon Application Recovery Controller (ARC) Region switch plan, and prove RTO 45 minutes and RPO 5 minutes in drills, not on paper.
- **Own:** each journey SLO has a named product owner; SRE owns the platform, the standards and the drills, under a graduated error budget policy that protects delivery capacity.

Assumptions and exclusions

#	Assumption or exclusion	Why it matters
A1	The critical data set for RTO and RPO is the transaction record in Amazon Aurora plus idempotency records in Amazon DynamoDB. Events published to Amazon MSK are written first to a transactional outbox table in Aurora, so Aurora replication also protects in-flight events.	Defines what the 5-minute RPO covers and removes MSK as a separate system of record.
A2	Primary and standby are two AWS Regions inside the same data residency boundary. The pair is confirmed against availability of Aurora Global Database, MSK Replicator and ARC Region switch before build.	Not every AWS feature is in every Region.
A3	Both external providers expose HTTPS APIs with contractual SLAs below 99.99% (assumed 99.9%), support idempotency keys, and allowlist caller IP addresses.	Drives fallback design and pre-registration of standby egress IPs.
A4	Provider calls are asynchronous to the customer: the customer receives "accepted" and fulfilment completes later.	Lets the synchronous journeys hold 99.99% even when a provider is below it.
E1	CI/CD design is out of scope except release gating by error budget and deployment of configuration during failover.	Scope control.
E2	Detailed cost modelling is out of scope; cost trade-offs are stated qualitatively.	Scope control.

AI assistance disclosure. This submission was prepared with two AI assistants: Claude (Anthropic, Claude Opus 5.5) and Gemini (Google). I selected AWS, supplied the assessor's scenario and constraints, provided the initial architecture diagram, set the review criteria and made the final design decisions. Gemini produced an earlier draft and review comments, including the flag that AWS App Mesh reached end of support on 30 September 2026. Claude drafted most of the final text, produced the architecture diagram, and verified service status and metrics against AWS documentation, which led to

replacing App Mesh with Istio on EKS, Systems Manager Incident Manager with CloudWatch and SNS alert routing, and ARC readiness checks with ARC Region switch plan evaluation. I cross-checked the two assistants against each other and against the cited sources (see References) before accepting any claim, and I take responsibility for the submitted content.

Where each required deliverable is

Required deliverable	Section	Task and weight
Service reliability model	Service reliability model (Task 1)	Task 1, 20%
SLI/SLO specification	SLI/SLO specification and error budget policy (Task 1)	Task 1, 20%
Observability and resilience architecture	Parts 1 and 2 (Tasks 2 and 3)	Task 2, 15%; Task 3, 20%
DR runbook and game-day plan	DR runbook and game-day plan (Task 4)	Task 4, 20%
Incident command model	Incident command model (Task 5)	Task 5, 15%
Prioritized 90-day roadmap	Prioritized 90-day roadmap (Task 6)	Task 6, 10%
Mandatory architecture diagram	Mandatory architecture diagram (AWS)	All tasks

Service reliability model (Task 1)

Three critical user journeys (CUJs) carry the 99.99% target; every dependency, failure mode and owner below is mapped to them.

Critical user journeys

ID	Journey	Path	Mode	Target
CUJ-1	Submit a transaction	Client → CloudFront + AWS WAF → API Gateway (REST, Regional) → VPC Link → internal NLB → EKS transaction service → RDS Proxy → Aurora writer; idempotency check in DynamoDB; event written to Aurora outbox	Synchronous	99.99% availability
CUJ-2	View transaction or account status	Client → CloudFront + AWS WAF → API Gateway → EKS query service → Aurora reader	Synchronous	99.99% availability
CUJ-3	Fulfil the transaction with external providers	Outbox relay → MSK topic → EKS consumer → Network Firewall egress → Provider 1 or 2 → result event → customer notified	Asynchronous	99.9% completed within 5 minutes

Dependency map

Dependency	Used by	Failure domain	How failure shows	Design response
CloudFront + AWS WAF	CUJ-1, 2	Global edge	Edge 5xx, origin timeouts	Origin resolved through a Route 53 failover record controlled by ARC; WAF rule changes staged in count mode first
API Gateway (REST, Regional)	CUJ-1, 2	Region	5xx, 429, latency	Method-level throttling, per-route integration timeouts, Region switch

Dependency	Used by	Failure domain	How failure shows	Design response
EKS services + Istio	CUJ-1, 2, 3	AZ, Region	Pod crash, saturation, sidecar misconfiguration	Three-AZ topology spread, PodDisruptionBudgets, ARC zonal shift, canary deploys
Aurora Global Database	CUJ-1, 2	AZ, Region	Writer failover, write stalls, replication lag	Multi-AZ writer and readers, RDS Proxy, Global Database secondary in standby Region
DynamoDB Global Tables	CUJ-1	Region	Throttling, replication delay	On-demand capacity, Aurora unique constraint as final duplicate guard
Amazon MSK	CUJ-3	AZ, Region	Broker loss, consumer lag, poison messages	3 AZs, replication factor 3, min.insync.replicas 2, retry and DLQ topics, MSK Replicator
Provider 1, Provider 2	CUJ-3	Outside AWS	Timeouts, error spikes, slow responses, full outage	Timeouts, circuit breakers, per-provider bulkheads, fallback, controlled queue drain

Failure modes

#	Failure mode	Impact	Detection	Primary mitigation	Owner
F1	Retry storm: a provider slows, callers hold connections, retries at several layers multiply load, pools exhaust and healthy paths fail (root cause of both past regional incidents)	Region-wide CUJ-1 and CUJ-2 failure	Retry ratio, pending requests per upstream, provider p99 latency, journey burn rate	One retry layer under a 20% budget, timeouts below caller deadlines, breakers, bulkheads	Service owner (config), SRE (mesh defaults)
F2	Bad deployment or configuration change (the most frequent cause of incidents in most services)	Any CUJ	Canary analysis on journey SLIs	Progressive delivery with automated rollback, error budget release gate	Service owner

#	Failure mode	Impact	Detection	Primary mitigation	Owner
F3	AZ impairment, including partial packet loss	Elevated errors in one AZ	Per-AZ SLIs, ARC zonal autoshift signals	ARC zonal shift, capacity sized to lose one AZ	SRE platform
F4	Aurora writer failure	CUJ-1 write errors during failover	RDS events, write error rate	Multi-AZ failover, RDS Proxy holds client connections through failover, client retry under budget	SRE platform
F5	MSK consumer lag or poison message	CUJ-3 delayed	Consumer lag, DLQ depth	Lag-based scaling, retry and DLQ topics, redrive runbook	Service owner
F6	Provider outage	CUJ-3 delayed	Breaker state, provider canary	Fallback provider or park in retry topic, controlled drain on recovery	Service owner + vendor manager
F7	Region impairment	All CUJs	Synthetics from two external vantage Regions, burn rate, AWS Health	Warm standby and ARC Region switch (Task 4)	SRE (execution), Incident Commander (decision)
F8	Thundering herd on recovery: clients and queues retry at once	Re-failure after recovery	Request rate vs baseline, 429 rate	API Gateway throttling with Retry-After, rate-limited queue drain	Service owner

Ownership

Each journey has one accountable product owner; SRE is accountable for the platform and the reliability standards, not for other teams' services.

Area	Product team	SRE	Platform engineering	Engineering leadership
Journey SLOs and error budget spend	A, R	C	I	I
Service code, timeouts, idempotency, fallbacks	A, R	C	I	I
Mesh resilience defaults and guardrails	C	A, R	R	I

Area	Product team	SRE	Platform engineering	Engineering leadership
EKS, MSK, Aurora operations	I	A	R	I
DR plan, Region switch drills	R (app validation)	A, R	R	I
Provider fallback and contracts	A, R	C	I	C
Error budget policy decisions	R	R	I	A

A = accountable, R = responsible, C = consulted, I = informed.

SLI/SLO specification and error budget policy (Task 1)

At 99.99% over a rolling 30 days the error budget is 4.32 minutes of full outage, or 1 failed request in 10,000. That budget is too small to absorb even one regional failover at the full 45-minute RTO (about 86% of the 52.6-minute annual budget), so availability must come from in-Region redundancy and fast detection, with Region switch as the last resort.

SLI and SLO table

SLI	Journey	Good event	Valid event	Source (AWS)	SLO, 30-day
Availability	CUJ-1	Response not 5xx and not 429	All requests except other client 4xx	CloudFront real-time logs via Kinesis Data Streams, aggregated per CUJ into CloudWatch metrics each minute (alerting); standard logs in S3 via Athena (monthly reporting)	99.99%
Latency	CUJ-1	Completed under 800 ms at the edge	All successful requests	CloudFront logs (time-taken)	99.9%

SLI	Journey	Good event	Valid event	Source (AWS)	SLO, 30-day
Availability	CUJ-2	Response not 5xx and not 429	As CUJ-1	CloudFront logs	99.99%
Latency	CUJ-2	Completed under 300 ms at the edge	All successful requests	CloudFront logs	99.9%
Synthetic journey success	CUJ-1, 2	Canary run passes all steps	All canary runs	CloudWatch Synthetics, both Regions, every minute	99.99% (corroborates low-traffic periods)
Fulfilment freshness	CUJ-3	Provider-confirmed within 5 minutes of acceptance	All accepted transactions	Outbox and result timestamps, metrics in Amazon Managed Service for Prometheus	99.9%
Recovery point (internal)	Critical data	AuroraGlobalDBRPOLag under 60 s	All 1-minute periods	Amazon CloudWatch, standby Region	99.9% of minutes
Provider health (dependency)	CUJ-3	Call succeeds within its timeout	All provider calls	Istio egress metrics in Amazon Managed Service for Prometheus	Tracked against each contract SLA; not a customer SLO

Design choices behind the table:

- **Measured at the edge.** API Gateway logs miss failures at CloudFront, DNS and WAF; customers do not.
- **429 counts as bad.** Load shedding protects the system but the customer still failed, so it spends budget honestly.
- **Latency is a separate 99.9% SLO.** A combined 99.99% "fast and successful" target would page on normal tail latency.
- **Providers get dependency SLIs, not customer SLOs.** A provider at 99.9% cannot carry a 99.99% journey, which is why CUJ-3 is asynchronous and has fallbacks.

Error budget policy

The policy is graduated so a single bad week redirects effort before it stops delivery. Freezes apply to the owning service only, never the whole organization.

Budget consumed (rolling 30 days)	Action	Effect on delivery
Under 50%	Normal operation; standing reliability allocation	About 80% feature work, 20% reliability work per team
50% to 75%	Reliability items move to the top of the owning team's backlog	Features continue through canary deployment
75% to 100%	Only low-risk changes ship (behind feature flags, off by default); open postmortem actions come first	Feature throughput reduced
Over 100%	Release freeze for the owning service except fixes and security patches	Lifted when the burn rate stays under 1.0 for 7 days and postmortem actions are scheduled

Two guardrails keep the policy fair: a freeze needs the budget exhausted on both the 30-day and 90-day windows, so one incident does not stop a quarter's delivery; and any incident that consumes over 25% of the 30-day budget requires a blameless postmortem within 5 business days. Exceptions need joint approval from the product director and the SRE lead, recorded in the decision log.

Observability and resilience architecture, part 1: observability (Task 2)

The goal is to cut time to detect customer impact to under 5 minutes while cutting pages: alerts fire on journey error budget burn, and component signals become diagnostics rather than pages.

Telemetry standards

- **One standard:** OpenTelemetry, collected by AWS Distro for OpenTelemetry (ADOT) running as an EKS add-on.
- **Required attributes on every signal:** service.name, service.version, deployment.environment, cloud.region, cloud.availability_zone, cuj.id, and the idempotency key on write paths.

- **Trace context end to end:** W3C traceparent from API Gateway into EKS, through Istio sidecars, into MSK message headers so consumers continue the same trace, and onto provider calls.
- **Logs:** structured JSON carrying trace_id and span_id, so any log line opens its trace.
- **Metrics:** RED (rate, errors, duration) per service, per route and per provider, plus saturation (pending requests, retry ratio, breaker state, consumer lag).

Signal pipeline, storage and retention

Signal	Collection	AWS store	Retention	Behaviour and trade-off
Service and mesh metrics	ADOT scraping app and Envoy endpoints	Amazon Managed Service for Prometheus	150 days default	High-cardinality PromQL without running Prometheus; cost grows with active series, so labels are reviewed
SLO aggregates	Recording rules and Athena jobs	Amazon CloudWatch metrics	15 months (rolled up)	Year-on-year SLO history at low cost; lower resolution after 15 days
Traces	ADOT with tail sampling: all errors and slow traces, 5% of the rest	AWS X-Ray	30 days	Keeps every failing trace while controlling cost; rare successful paths may be unsampled
Application logs	Fluent Bit on EKS	CloudWatch Logs, exported to S3	30 days hot, 1 year in S3 with lifecycle to archive	Fast queries for recent incidents, cheap audit retention
Edge logs	CloudFront standard logs	S3, queried with Athena	13 months	Source of the journey SLIs

Signal	Collection	AWS store	Retention	Behaviour and trade-off
Change events	CloudTrail organization trail, EventBridge deployment events	S3 and CloudWatch	1 year minimum	Correlates incidents with changes (failure mode F2)

Correlation and service maps

- **Service map:** CloudWatch Application Signals and the X-Ray trace map show EKS services, Aurora, DynamoDB, MSK and provider calls with live error and latency, so responders see where a failure is spreading.
- **Journey dashboards:** Amazon Managed Grafana joins the CloudFront SLIs, Prometheus metrics, X-Ray traces and business signals on one page per CUJ, with deployment markers from EventBridge.
- **Trade-off:** two consoles (Application Signals for services, Grafana for journeys) is one more tool, accepted because Grafana is the only place the edge SLIs and business signals meet.

Synthetic checks

Canary (CloudWatch Synthetics)	What it does	Frequency	Runs from
CUJ-1 write	Submits a test transaction for a synthetic tenant, with an idempotency key, through the public CloudFront endpoint	1 minute	Two Regions other than the primary
CUJ-2 read	Reads the synthetic transaction status	1 minute	Same
Standby origin	Calls the standby Region's API Gateway directly, bypassing the failover record	1 minute	Primary and a third Region
Provider sandbox	Calls each provider's sandbox endpoint through Network Firewall egress	5 minutes	Each Region

The standby origin canary is what makes DR readiness continuous instead of quarterly. Synthetic tenants are excluded from business signals so tests do not distort revenue metrics.

Business signals

Transactions accepted per minute, fulfilment completions per minute and provider confirmation rate are emitted as CloudWatch metrics using the embedded metric format. CloudWatch anomaly detection compares each against the same time last week, which catches failures that return HTTP 200, such as a silent drop in completions.

Alert routing

Pages fire on multi-window, multi-burn-rate alerts against each journey SLO, following the Google SRE workbook method:

Alert	Condition (99.99% SLO)	Budget consumed	Action
Fast burn	Burn rate 14.4 over 1 hour and 5 minutes (error rate above 0.144%)	2% in 1 hour	Page product on-call, SRE secondary
Slow burn	Burn rate 6 over 6 hours and 30 minutes	5% in 6 hours	Page product on-call
Budget trend	Burn rate 1 over 3 days and 6 hours	10% in 3 days	Ticket to the owning team
Platform saturation	Breaker open, consumer lag, RPO lag above 60 s	Not budget based	Page SRE platform on-call if it threatens an SLO, otherwise ticket

Routing path: Amazon Managed Service for Prometheus alert manager and CloudWatch alarms publish to Amazon SNS, which feeds the organization's paging tool; AWS Health events reach the incident channel through EventBridge. AWS Systems Manager Incident Manager is not used because it closed to new customers on 7 November 2025. Cause-based alerts such as CPU, memory and pod restarts are demoted to dashboards or tickets, with a target of at least 50% fewer pages and at least 80% of pages actionable within 30 days.

Observability and resilience architecture, part 2: resilience (Task 3)

The two regional incidents were retry storms, so the design rule is that load from a failing dependency must stay bounded: one retry layer, a 20% retry budget, deadlines that shrink at each hop, and shedding before saturation.

Service mesh choice: Istio on EKS

AWS App Mesh reached end of support on 30 September 2026 (AWS notice), so it is not an option. AWS positions Amazon VPC Lattice as a migration path (AWS blog); Lattice removes sidecars and adds IAM-based service authorization, but it does not provide retry budgets or outlier detection, which are the controls failure mode F1 needs.

Istio runs on EKS with Envoy sidecars and integrates with AWS-native identity (IAM Roles for Service Accounts), telemetry (ADOT to Amazon Managed Service for Prometheus and X-Ray) and load balancing (AWS Load Balancer Controller). The trade-off is that the team operates the Istio control plane: upgrades are kept within one minor version of upstream, rehearsed in staging, and sidecars add CPU and memory overhead per pod. Istio ambient mode, which removes sidecars, is reviewed after the 90 days. VPC Lattice remains an option for future cross-VPC or cross-account service connectivity, as a complement rather than a replacement.

Controls by layer

Control	Where	Setting	Behaviour	Trade-off
Timeouts and deadlines	API Gateway integration, Istio VirtualService, app clients, Aurora	CUJ-1: API Gateway 5 s, service route 4 s, Aurora statement 2 s, provider call 2 s; remaining deadline propagated in a request header	Each hop gives up before its caller does, so no work continues for a caller that has gone	Tight timeouts fail slow-but-valid requests; values set from measured p99.9 plus margin and reviewed monthly
Retries	Caller's Istio sidecar only; library and SDK retries disabled for service-to-service calls	Up to 2 retries on connect failure, reset, 503; jittered backoff 25 ms to 250 ms; non-idempotent requests retried only with an idempotency key	One layer means retries add, never multiply	Developers lose control of retries in code; documented as a platform standard

Control	Where	Setting	Behaviour	Trade-off
Retry budget	Envoy cluster circuit_breakers.retry_budget, applied with an Istio EnvoyFilter (VirtualService sets attempts but not budgets)	budget_percent 20, min_retry_concurrency 3	Once retries exceed 20% of active requests, further retries fail fast	EnvoyFilter is version-sensitive; covered by an upgrade test in staging
Retries to AWS APIs	AWS SDKs	Adaptive retry mode, 3 attempts	Client-side rate limiting backs off under throttling	Separate dependency from service calls, so not a stacked layer
Circuit breaking	Istio DestinationRule outlier detection	5 consecutive 5xx, 10 s interval, 30 s base ejection, at most 50% of endpoints ejected	Unhealthy endpoints leave the pool; the 50% cap avoids ejecting everything	A noisy but healthy pod can be ejected briefly
Bulkheads	Istio DestinationRule connection pools, one per provider and per critical upstream	Max connections, pending requests and concurrent requests per provider	A slow provider exhausts only its own pool, not the service	Limits must be re-tuned when traffic grows
Provider fallback	Application, with breaker state from the mesh	Interchangeable providers fail over to each other; otherwise work is parked in an MSK retry topic and the customer sees "pending"	CUJ-1 still returns success while CUJ-3 degrades gracefully	Fulfilment delay counts against the CUJ-3 freshness SLO

Control	Where	Setting	Behaviour	Trade-off
Idempotency	API Gateway requires an Idempotency-Key header on POST and PUT; DynamoDB conditional write (attribute_not_exists) with 24-hour TTL; Aurora unique constraint on client transaction ID	Keys also forwarded to providers that support them	Retries, client resubmits and replays cannot create duplicates	DynamoDB Global Tables are last-writer-wins, so after an unplanned Region switch the Aurora constraint is the final guard and reconciliation covers provider-side duplicates
Backpressure	Envoy pending-request limits, MSK consumers	Consumers pause partitions while a provider breaker is open; scaling on consumer lag (KEDA reading Amazon Managed Service for Prometheus) capped at each provider's rate limit	Load waits in MSK, where it is durable, instead of in memory	Fulfilment latency rises during provider trouble
Queue recovery	MSK retry topics and DLQ topic	Retry after 1, 5 and 30 minutes; DLQ after 3 attempts; topic retention 7 days; producer idempotence on; offsets committed after processing	Poison messages leave the main path; a redrive runbook replays the DLQ at a controlled rate after a fix	DLQ needs an owner and a daily check
Load shedding	AWS WAF rate-based rules; API Gateway usage plans and method throttling (token bucket rate plus burst) returning 429 with Retry-After; AWS AppConfig feature flags	CUJ-2 and non-critical routes throttled before CUJ-1 writes; brownout flags switch off non-essential features	Overload becomes fast rejections instead of slow failures for everyone	Rejected requests still spend error budget, by design

Control	Where	Setting	Behaviour	Trade-off
Outbox relay	Aurora outbox table, relay service to MSK	Event and transaction committed together	No lost or phantom events when MSK or the relay fails	Extra write per transaction

Capacity strategy

- **Survive one AZ:** each tier runs across 3 AZs with enough capacity to lose one at peak, which means about 150% of peak in total. HPA minimum replicas are sized for this, and topology spread constraints keep pods even across AZs.
- **Compute:** Karpenter provisions EKS nodes across several instance types to reduce the risk of capacity shortfall in one type.
- **Data:** Aurora writer plus two readers across 3 AZs, kept under 60% CPU at peak; MSK with brokers spread evenly across 3 AZs, replication factor 3, min.insync.replicas 2 and storage auto scaling; DynamoDB on-demand.
- **Standby Region:** compute at 20% of production, with On-Demand Capacity Reservations for the node count needed to reach 100%, so scale-out does not compete for capacity while many customers fail over at once. The data tier runs at full size because the Aurora secondary must carry production load the moment it is promoted.
- **AZ impairment:** ARC zonal shift (with zonal autoshift enabled) moves traffic away from an impaired AZ for the load balancer and EKS, without a Region switch.
- **Verification:** quarterly load tests with the Distributed Load Testing on AWS solution at 2x forecast peak, plus scale-out timing in every DR drill.

DR runbook and game-day plan (Task 4: regional recovery)

Recovery uses a warm standby Region and a scripted ARC Region switch plan, targeting 30 minutes from decision to recovered traffic so the 45-minute RTO holds with margin, and an RPO measured in seconds against a 5-minute limit.

Strategy and trade-offs

Option	RTO	RPO	Why chosen or rejected
Pilot light	1 hour or more	Seconds	Rejected: building compute during an event risks the 45-minute RTO

Option	RTO	RPO	Why chosen or rejected
Warm standby (chosen)	Under 45 minutes	Seconds	Compute at 20% scales out in minutes; data tier already replicated at full size
Active-active	Minutes	Near zero	Rejected for now: Aurora has one writer, so active-active writes need application-level conflict handling that would consume most of the 90-day delivery capacity

Failure detection and authority

- **Signals that start the clock:** fast-burn alerts on CUJ-1 or CUJ-2; CloudWatch Synthetics failing from two external vantage Regions; AWS Health events for the primary Region.
- **First response is in-Region:** ARC zonal shift, rollback of recent changes and load shedding. A Region switch is considered when these have not restored service within 15 minutes, or when impact is projected to exceed 30 minutes.
- **Authority:** the Incident Commander recommends; one named approver (Head of SRE, with two named delegates) authorizes. If no approver answers within 5 minutes during a SEV-1, the Incident Commander is pre-authorized to proceed. The decision and its timestamp go in the decision log.
- **Readiness:** ARC Region switch plan evaluation runs continuously and the standby origin canary runs every minute. ARC readiness checks are not used because they are closed to new customers and AWS recommends Region switch plan evaluation instead (AWS notice).

DR runbook: unplanned Region switch

Step	Target time	Action	Evidence
0	T+0	Detection: fast-burn page	Alarm history
1	T+5	Incident Commander declares SEV-1 and starts the decision clock	Decision log
2	T+15	Decision point: approver authorizes the ARC Region switch plan, executed from outside the impaired Region	Region switch execution record

Step	Target time	Action	Evidence
3	T+15	Pre-checks: plan evaluation green, standby canary green; record current AuroraGlobalDBRPOLag as the expected data loss window	Screenshot and metric value in log
4	T+16	Stop writes in the primary if it is reachable: pause outbox relay and MSK consumers, set API Gateway maintenance responses through AppConfig	AppConfig deployment ID
5	T+17	Scale standby EKS to 100% (raise HPA minimums; Karpenter uses the capacity reservations)	Node and pod counts
6	T+18	Promote data: if the primary is reachable, Aurora Global Database switchover (no data loss); if not, managed failover with data loss accepted, which promotes the standby cluster to writer	RDS event IDs
7	T+25	Messaging: standby consumers start on the MSK Replicator target cluster using translated consumer group offsets; producers repointed through AppConfig	Consumer lag graph
8	T+25	DynamoDB Global Tables are already writable in the standby Region; confirm writes succeed	Test write
9	T+27	Dependency validation (list below)	Checklist signed in log
10	T+30	Traffic shift: ARC routing control switches the Route 53 api-origin record to the standby API Gateway; CloudFront keeps serving with a 60-second origin DNS TTL	Routing control state change
11	T+35	Verify: synthetic canaries pass, burn rate falling, business signals recovering	Dashboard snapshot
12	T+45	RTO limit. Declare recovered; record measured RTO and RPO	Incident timeline

Dependency validation checklist

- Provider 1 and Provider 2 allowlist the standby Region's NAT Gateway Elastic IPs (registered in advance and re-tested in every drill).
- TLS certificates for the API and origin exist in AWS Certificate Manager in the standby Region.
- Secrets Manager replica secrets and the KMS multi-Region key decrypt correctly in the standby Region.
- Network Firewall FQDN allowlist in the standby Region matches the primary (managed as code).
- WAF web ACL, API Gateway throttles and authorizer configuration match the primary (deployed from the same templates).
- Provider sandbox canary passes from the standby Region.

Reconciliation after an unplanned switch

Transactions committed in the primary during the replication lag window (normally seconds, alarmed at 60 seconds, limited to 5 minutes) may be missing from the promoted database.

1. Preserve the former primary's data before it is rebuilt as a secondary: snapshot it as soon as the Region is reachable.
2. Compare that snapshot with the new primary by client transaction ID over the lag window.
3. Replay missing transactions through the normal API using their original idempotency keys, so nothing is applied twice.
4. Compare provider records (provider confirmation IDs) with local records to find work that reached a provider but not the database; reverse or confirm with the provider.
5. Contact affected customers if any transaction outcome changed, and record totals in the postmortem.

Failback

Failback is a planned change in a low-traffic window, at least 24 hours after the original Region is healthy:

1. Re-add the original Region to the Aurora Global Database as a secondary and wait until replication lag is under 1 second.
2. Reverse MSK Replicator direction and confirm consumer offsets are in sync.
3. Scale the original Region's EKS to 100%.

4. Run Aurora Global Database switchover (no data loss), then the Region switch plan in reverse.
5. Return the former standby to 20% compute after 24 hours of stable SLIs.

Writes run in one Region at a time, so traffic moves back in one step after switchover rather than gradually; canaries and burn-rate alerts gate each step.

Game-day plan

Every experiment runs through AWS Fault Injection Service (FIS) with CloudWatch alarm stop conditions tied to the journey burn rate, so an experiment halts itself if real customers start to suffer.

Game day	Method	Environment and frequency	Pass criteria
AZ loss	FIS AZ availability power interruption scenario	Staging monthly; production quarterly	ARC zonal shift completes, CUJ SLIs stay within SLO, no page above SEV-3
Retry storm reproduction	Istio fault injection adds 2 s latency to a provider at 2x peak load	Staging monthly until it passes 3 times, then quarterly	Retries stay at or under 20% of requests, breakers open within 30 s, other services unaffected
Provider outage	Network Firewall rule drops traffic to one provider	Staging monthly; production quarterly in low traffic	Fallback or parking works, CUJ-1 stays within SLO, queue drains at the provider rate limit after recovery
Full Region switch	ARC Region switch plan executed end to end, including reconciliation	Staging twice in Month 3; first production run at day 90 in a low-traffic window, then quarterly	Measured RTO at or under 45 minutes (target 30), measured RPO at or under 5 minutes, zero duplicate transactions after reconciliation
Failback	Planned switchover back	After each Region switch drill	Completed with no data loss and no SLO breach

Game day	Method	Environment and frequency	Pass criteria
Decision tabletop	Facilitated exercise with the Incident Commander pool and approvers	Quarterly	Decision made within 15 minutes of the scenario start

Each drill produces evidence for the assessment trail: the FIS experiment report, the Region switch execution record, measured RTO and RPO, and a short findings report whose actions enter the roadmap backlog.

Incident command model (Task 5)

Severity is set by customer impact measured against the error budget, not by which component broke, so a SEV-1 is declared within 5 minutes of a fast burn instead of after a fixed outage duration.

Severity criteria

Severity	Criteria (any one)	Response	Communication cadence
SEV-1	Fast-burn alert on CUJ-1 or CUJ-2 sustained for 5 minutes; journey success under 99%; suspected data loss or integrity risk; Region switch under consideration	Page Incident Commander, owning team and SRE within 5 minutes; executive liaison informed within 15 minutes	Status page and stakeholders every 15 minutes; executive summary every 30 minutes
SEV-2	Slow-burn alert; partial degradation of one journey; CUJ-3 freshness SLO breached; one provider down with fallback active	Page owning team on-call; Incident Commander joins on request	Stakeholders every 60 minutes
SEV-3	Redundancy lost with SLIs intact: AZ shifted away, breaker open with fallback holding, DLQ growing	Ticket for the owning team, handled in business hours	Daily update in the ticket

Any responder can raise the severity; only the Incident Commander lowers it, with the reason in the decision log.

Roles

Role	Filled by	Responsibilities
Incident Commander (IC)	Trained rotation drawn from SRE and senior product engineers	Owns the incident, sets priorities, makes or escalates decisions including Region switch recommendation; does not debug
Operations lead	Owning team's on-call engineer	Directs technical investigation and mitigation; runs runbooks
Communications lead	Product or support duty manager	Status page, customer and stakeholder updates on the cadence above

Role	Filled by	Responsibilities
Scribe	Any trained engineer not on operations	Keeps the timeline and decision log in real time
Provider liaison	Vendor manager or owning team lead	Opens and tracks provider escalations under contract terms
Executive liaison (SEV-1)	Engineering director on call	Business decisions, customer commitments, Region switch approval if delegated

Decision log and evidence

- **Channel:** a bot opens one incident channel per incident in the organization's chat tool and pins the timeline document.
- **Decision log:** each entry records time, decision, who decided, options considered and the evidence used. Region switch decisions also record the approver and the measured AuroraGlobalDBRPOLag at the time.
- **Immutable record:** chat messages can be edited, so at close the timeline, decision log and channel export are written to an Amazon S3 bucket with S3 Object Lock in compliance mode, retained for at least one year.
- **Evidence collected automatically:** CloudWatch alarm history, Grafana dashboard snapshots, X-Ray traces for failing requests, CloudTrail change events, ARC routing control and Region switch execution history, FIS experiment reports, and AWS Health events.

Escalation

Trigger	Escalate to
Page not acknowledged in 5 minutes	Secondary on-call, then the team's engineering manager at 10 minutes
SEV-1 unresolved at 30 minutes	Head of Engineering and executive liaison
AWS service suspected	AWS Support case at the highest severity the support plan allows, with the incident ID
Provider suspected	Provider's contractual escalation path, opened by the provider liaison
Region switch considered	Named approver per Task 4, with the pre-authorization rule after 5 minutes

Handover

Handover happens at shift change or when an IC has led for 4 hours. The outgoing IC writes a handover note in the timeline (current impact, active hypotheses, actions in flight, next decision and its deadline, open escalations), then holds a 10-minute call with the incoming IC. The incoming IC posts "I have command" in the channel and the scribe logs the time.

After the incident

SEV-1 and SEV-2 incidents, and any incident that spends over 25% of a 30-day budget, get a blameless postmortem within 5 business days. Actions are tracked in the owning team's backlog with an owner and due date, and overdue actions are reviewed in the monthly reliability review (Task 6).

Prioritized 90-day roadmap (Task 6)

The 90 days are ordered by risk to the journey SLOs: Month 1 fixes detection and stops retry amplification, Month 2 builds platform resilience, Month 3 proves recovery. Product teams commit 20% of sprint capacity to roadmap items, so feature delivery continues at about 80%.

How items are prioritized

Each item is scored before it enters the backlog:

- **Risk score:** likelihood (incidents in the last 12 months, game-day findings) × impact (estimated error budget minutes lost per occurrence). The two retry-storm incidents and the untested DR path score highest.
- **Toil saved:** engineer hours per week removed, from pager data (pages per week, share actionable) and a two-week toil log kept by SRE and on-call engineers in weeks 1 and 2.
- **Order:** highest risk first; among similar risks, highest toil saved first.

Backlog

Month	Item	Problem fixed	Risk	Toil saved	Owner
1	Name an owner per CUJ, sign the RACI and the error budget policy	Ownership	High	Low	Engineering leadership
1	Publish CUJ SLIs and SLOs; build edge SLI pipeline and Grafana journey dashboards	Detection	High	Medium	SRE + product teams
1	Burn-rate alerts live; demote cause-based alerts	Detection, alert noise	High	High	SRE
1	CloudWatch Synthetics canaries for CUJ-1, CUJ-2, standby origin and providers	Detection, DR readiness	High	Low	SRE
1	Retry hotfix: disable stacked library retries, set timeouts, API Gateway throttles with Retry-After	Retry storms	Very high	Medium	Product teams
1	Baseline metrics: MTTD, MTTR, pages per week, toil hours	Governance	Medium	Low	SRE
2	Istio rollout to all services with retry budgets, outlier detection and per-provider bulkheads	Retry storms	Very high	Medium	SRE platform
2	Idempotency keys on all writes; Aurora outbox; MSK retry and DLQ topics with redrive runbook	Data integrity, queue recovery	High	High	Product teams
2	Provider fallbacks and AppConfig brownout flags	External dependencies	High	Low	Product teams
2	Retry storm game day in staging (FIS and Istio fault injection)	Retry storms	High	Low	SRE

Month	Item	Problem fixed	Risk	Toil saved	Owner
2	Start standby Region build as code: VPC, EKS at 20%, Aurora Global Database, MSK Replicator, DynamoDB replica	DR	Very high	Low	Platform engineering
3	Complete standby Region; dependency validation (provider allowlists, certificates, secrets)	DR	Very high	Low	Platform engineering
3	ARC Region switch plan with plan evaluation; reconciliation tooling	DR	Very high	Medium	SRE
3	Two staging Region switch drills plus failback; fix findings	DR	Very high	Low	SRE + product teams
3	First production Region switch drill at day 90 in a low-traffic window, after two clean staging runs	DR	Very high	Low	SRE, approved by engineering leadership

Day-90 success metrics

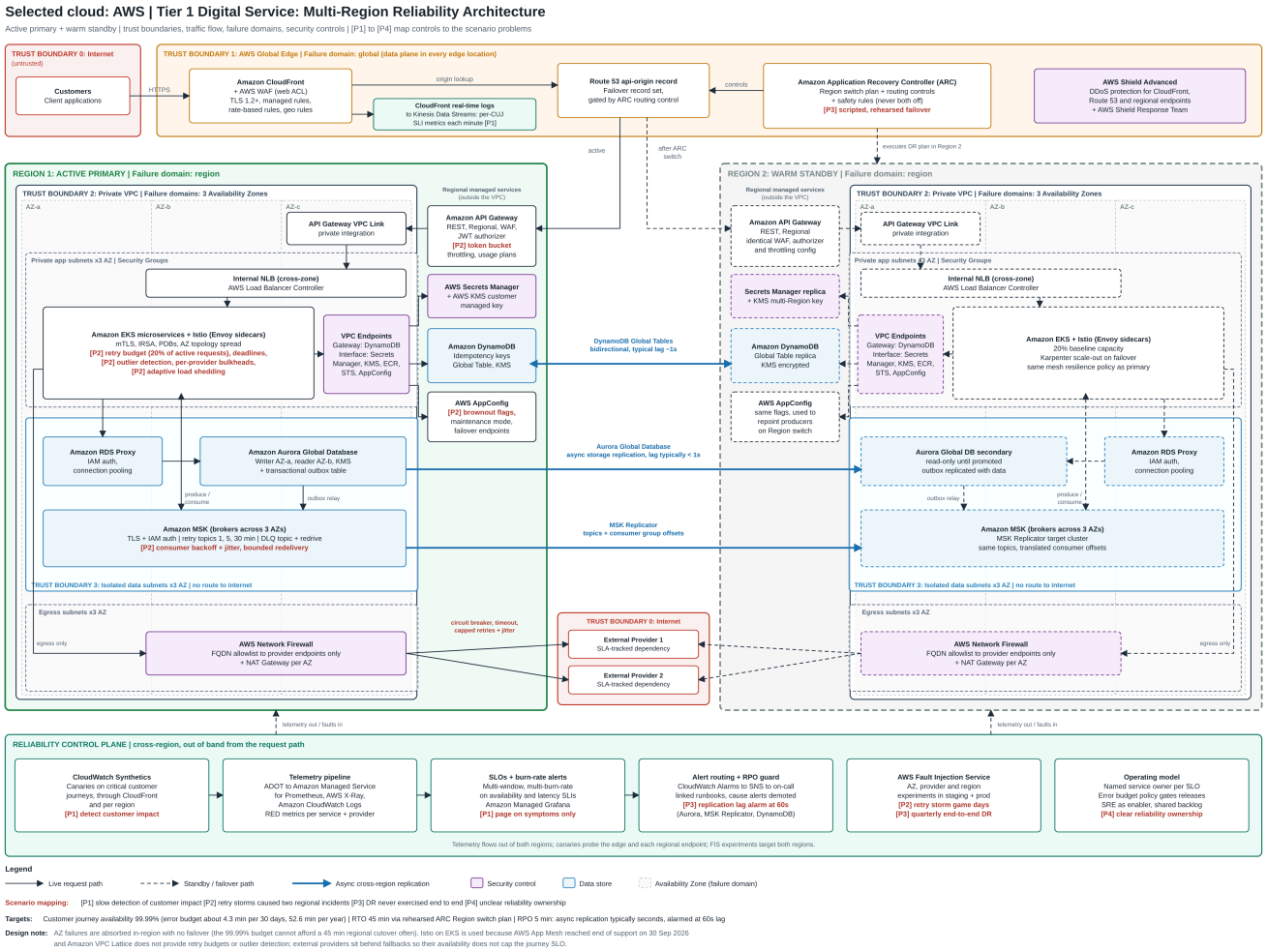
Metric	Baseline	Day-90 target
Time to detect customer-impacting failure	Measured in weeks 1 and 2	Under 5 minutes
Pages per on-call week	Measured in weeks 1 and 2	At least 50% fewer; at least 80% actionable
Retry share of requests during a provider slowdown	Unbounded (caused two incidents)	At or under 20%, shown in game day
Measured RTO and RPO	Never tested	RTO at or under 45 minutes (target 30), RPO at or under 5 minutes, in 2 staging and 1 production drill
CUJs with SLO, owner and error budget tracking	0 of 3	3 of 3
SRE time on operational work	Measured in weeks 1 and 2	At or under 50%, trending to 40%

Governance: keeping SRE from becoming an operations dumping ground

- **Engagement model:** SRE supports a service only after a production readiness review: an SLO, an owner, runbooks, dashboards and alerts that pass the burn-rate standard. Until then the product team carries the pager alone.
- **Pager ownership:** product teams are primary on-call for their services; SRE is primary for the shared platform (EKS, Istio, MSK, Aurora) and secondary for services.
- **Toil cap:** SRE operational work is capped at 50% of team time, measured weekly from tickets and pages. If the cap is exceeded for 2 consecutive weeks, the excess work goes back to the owning team until it is automated.
- **Recurring work rule:** any manual task done more than twice a month gets an automation ticket, or is returned to its owner.
- **Monthly reliability review:** product, SRE and engineering leadership review budget spend per CUJ, top risks, toil trend and overdue postmortem actions, and re-rank the backlog.
- **Quarterly reset:** the risk and toil scores are recalculated from the quarter's incidents and game days, and the next 90-day plan is agreed with product leadership so reliability work competes openly with features.

Mandatory architecture diagram (AWS)

The diagram shows the AWS reference workload as one active primary Region and one warm standby Region, with every control from Tasks 2 to 4 placed where it acts and tagged [P1] to [P4] against the scenario problems.



A full-resolution copy (4800 x 3600 px) is attached separately as tier1-aws-architecture.png so the labels stay readable when the document is printed or exported to PDF.

Requirement	Where it appears on the diagram
Trust boundaries	Boundary 0: internet (customers and both providers); Boundary 1: AWS global edge; Boundary 2: private VPC per Region; Boundary 3: isolated data subnets with no route to the internet
Traffic flow	Solid arrows: live path CloudFront + WAF → Route 53 origin record → API Gateway → VPC Link → internal NLB → EKS → RDS Proxy, Aurora, MSK, DynamoDB; egress only through Network Firewall; outbox relay from Aurora to MSK; MSK retry and DLQ topics; CloudFront real-time logs to Kinesis Data Streams for edge SLIs. Dashed arrows: standby path after an ARC switch. Thick blue arrows: asynchronous replication

Requirement	Where it appears on the diagram
Failure domains	Global edge, each Region, and three Availability Zone lanes inside each VPC
Security controls	AWS Shield Advanced, AWS WAF, TLS 1.2+, JWT authorizer, private VPC Link, Security Groups, Istio mTLS, IAM Roles for Service Accounts, RDS Proxy IAM auth, MSK TLS + IAM, KMS encryption, Secrets Manager, VPC endpoints, Network Firewall FQDN allowlist
Key service names	CloudFront, AWS WAF, Route 53, ARC Region switch and routing controls, API Gateway, EKS, Istio, NLB, Aurora Global Database, RDS Proxy, Amazon MSK and MSK Replicator, DynamoDB Global Tables, CloudWatch Synthetics, ADOT, Amazon Managed Service for Prometheus, X-Ray, Managed Grafana, AWS Fault Injection Service, AWS AppConfig, Kinesis Data Streams (CloudFront real-time logs)

Scenario tags: [P1] slow detection of customer impact, [P2] retry storms, [P3] DR never exercised, [P4] unclear reliability ownership.

Validation, limitations and references

How each claim is validated

Claim	Evidence	AWS source of evidence
99.99% journey availability	Monthly SLO report per CUJ	CloudFront logs aggregated into CloudWatch metrics; Grafana SLO dashboards
Detection under 5 minutes	Time from first failed request to page, per incident	CloudFront real-time logs vs SNS publish time
RPO under 5 minutes	Continuous lag record and drill measurements	AuroraGlobalDBRPOLag in the standby Region (AWS metric reference)
RTO under 45 minutes	Measured time from decision to recovered traffic in drills	ARC Region switch execution history, synthetic canary results

Claim	Evidence	AWS source of evidence
Retry storms contained	Retry share stays at or under 20% under injected provider latency	FIS experiment reports, Istio metrics in Amazon Managed Service for Prometheus
Load shedding works	Throttled requests return 429 quickly at 2x peak	Distributed Load Testing on AWS runs, API Gateway metrics

Known limitations

- **Cost:** the standby Region runs compute at 20% plus a full-size data tier and capacity reservations, an estimated third or more of production infrastructure cost (a rough estimate, to be confirmed with the AWS Pricing Calculator). Cost controls already in the design are tail-based trace sampling, Prometheus label review, log lifecycle to archive, and a 20% compute baseline instead of full duplication.
- **Unplanned failover loses the replication lag window:** normally seconds, always under the 5-minute RPO, but reconciliation (Task 4) is still required.
- **A regional failover is expensive against 99.99%:** one switch at the full 45-minute RTO uses about 86% of the annual error budget, so the design depends on in-Region resilience and fast detection.
- **Istio is customer-operated:** it is not an AWS managed service, so upgrades and sidecar overhead are the team's responsibility (Task 3 explains why it is still the right choice).
- **Providers bound CUJ-3:** fulfilment freshness cannot be better than the providers allow; fallbacks reduce but do not remove this.
- **Region pair:** service availability for Aurora Global Database, MSK Replicator and ARC Region switch must be confirmed for the chosen Regions before build.

References

- AWS App Mesh end of support notice
- Migrating from AWS App Mesh to Amazon VPC Lattice (AWS Containers blog)
- Getting started with Istio on Amazon EKS (AWS Open Source blog)
- ARC readiness check availability change and Region switch recommendation
- AWS Systems Manager Incident Manager availability change
- Amazon CloudWatch metrics for Amazon Aurora
- Alerting on SLOs, Google SRE workbook
- Region switch in ARC (AWS documentation)
- Choosing a Service Mesh on EKS (third-party comparison of VPC Lattice and Istio)

- AI assistance: Claude (Anthropic) and Gemini (Google), as disclosed under Assumptions and exclusions